JM RECORDS

Vocal Channel

User Manual

Version 1.0 (working name) · AU · VST3 · Standalone · JM Records



One plugin for the whole vocal mix: LEARN, 12.47 million vocal types, EQ, de-esser, compressors, saturation. Zero latency.

1. What this plugin is

JM Vocal Channel is **one plugin that handles the whole tonal and dynamic side of a vocal mix**: EQ, de-esser, a three-stage compressor chain (leveler, FET-style, opto-style plus a catch compressor), saturation and a colour stage. Reverb and delay are deliberately not inside: put them on a bus, as usual.

The idea is simple: **press LEARN, sing or play the vocal, pick the kind of vocal it is, and you are done.** The plugin listens to your real voice (how loud, how wide the dynamics are, how dark or bright it is, how much room noise and rumble there is, how sharp the S sounds are), and then calculates every setting in the signal path for the vocal type you picked. Afterwards you may open ADVANCED and make small corrections.

HOW IT WORKS IN ONE LINE

- **LEARN** measures your voice. It never changes the sound by itself and it works with a small coach that tells you what to sing or say next.
- **Vocal type** = a choice from eight lists (Style, Voice, Range, Source, Role, Era, Intensity, Feel). Together they make **12,474,000 different starting points**.
- One click on a type solves all stages at once, taking the LEARN result into account.
- **ADVANCED** shows the EQ, high-pass, saturation and input values the type has set, so you can nudge them.

HONEST NOTES

- The signal path is analog-modelled: **no look-ahead, no FFT in the audio path, zero latency**. It can be used on a live input.
- No claim is made that the plugin matches any hardware unit. Numbers inside are starting points, measured on nine real dry vocals; some are marked as assumptions in the source code.
- The 12.47 million types are **generated from rules**, not hand-voiced one by one. They are good starting points; your ears decide the final setting.

2. Install and open

Format	Where to copy it
Audio Unit (AU) — macOS only	`~/Library/Audio/Plug-Ins/Components/JM Vocal Channel.component`
VST3 — macOS	`~/Library/Audio/Plug-Ins/VST3/JM Vocal Channel.vst3`
VST3 — Windows (64-bit)	`C:\Program Files\Common Files\VST3\JM Vocal Channel.vst3` (the whole folder, with everything inside it)
Standalone	macOS: `JM Vocal Channel.app`; Windows: `JM Vocal Channel.exe`. Own window, uses the system audio device.

macOS: Audio Unit and VST3 plugins are 64-bit; the shipped build runs on Apple Silicon Macs (M1 or newer). **Windows:** the plugin is a 64-bit VST3 and needs a 64-bit DAW. The Audio Unit format does not exist on Windows.

1. Copy the files to the folders above. On Windows the `.vst3` is a **folder**: copy the whole folder, not only a file from inside it. Windows asks for administrator permission when it writes to `C:\Program Files`: choose Continue.
2. **macOS:** if the DAW does not show the plugin, quit the DAW, open Terminal and run `killall -9 AudioComponentRegistrar`, then rescan plugins in the DAW. If macOS says the plugin is damaged or from an unidentified developer (the build is not notarized), run `xattr -dr com.apple.quarantine "<path of the plugin>"` in Terminal once.
3. **Windows:** open the DAW and rescan plugins (many DAWs do it by themselves at start-up). Windows SmartScreen or an antivirus program may warn about a file that is not code-signed: if you trust the source, choose More info and then Run anyway, or allow the file. If the plugin does not load on a computer that never had other plugins installed, install the *Microsoft Visual C++ Redistributable (x64)* from Microsoft.
4. Insert it on a vocal track (mono or stereo). Put it **first** in the insert chain, before any reverb or delay. Mono tracks and stereo tracks both work.
5. The plugin reports **0 samples of latency**, so no delay compensation is needed.

Product name: JM Vocal Channel is a working name; the final product name is still open.

3. Quick start (four steps)

The plugin does **nothing** when you load it: the sound passes unchanged. It opens on a **setup screen**.

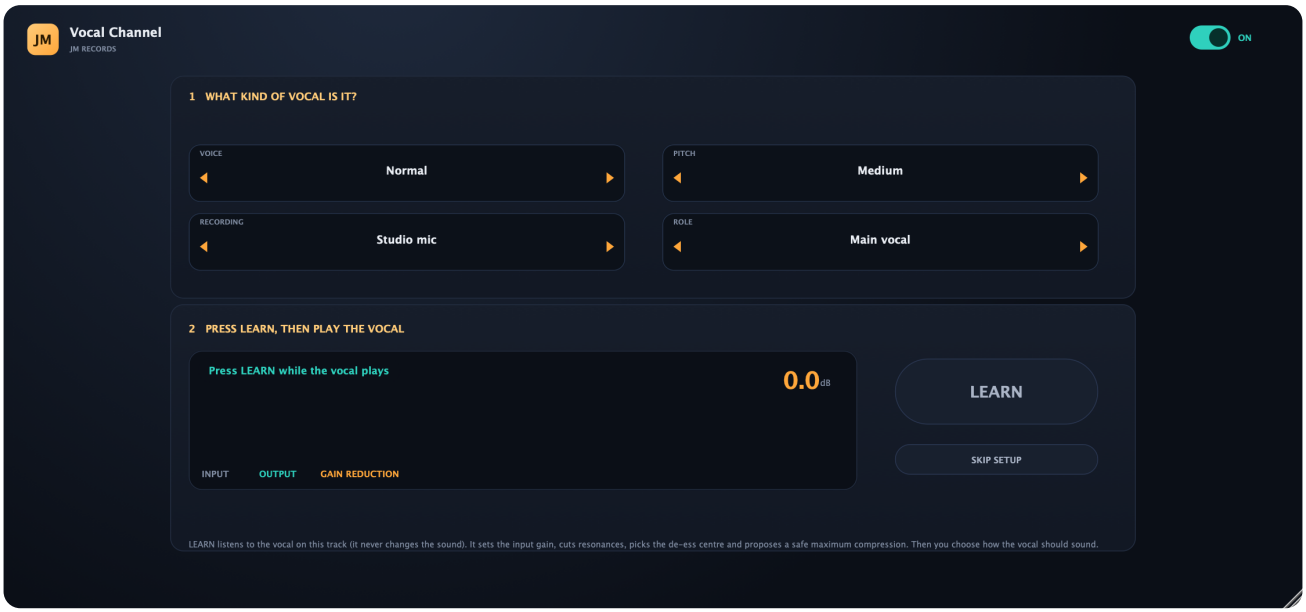
1. **Say what kind of vocal it is** (setup screen, step 1): VOICE, RANGE, SOURCE (which mic / room) and ROLE (lead, backing…).
2. **Press LEARN and play the recorded vocal** (step 2). The vocal that is already on this mixer / DAW track passes through the plugin and it listens. When LEARN is finished the main screen opens by itself.
3. **Pick a sound** in the list on the left. Optionally set COLOUR, STRENGTH and FEEL.
4. **Listen and trim**: MAX COMPRESSION, LOW, HIGH, MIX, OUTPUT. Everything else is under ADVANCED.

What LEARN does on its own (before you pick any style): it sets the INPUT gain, applies a first corrective EQ (MATCH: cuts resonances and harsh or boxy areas, high-passes rumble), chooses the de-ess centre and proposes a safe MAX COMPRESSION. **What the style does**: the colouring (EQ shaping, compression, saturation). Every style is set to the **same loudness** as the corrected, unstyled vocal, so you can compare styles fairly.

SKIP SETUP (bottom right of the setup screen) opens the main screen without LEARN; then the plugin uses average-voice values and cannot fit the compression limit to your vocal. The SETUP button (bottom row of the main screen) returns to the setup screen.

4. The two screens

Setup screen



Setup screen: choose the vocal, then LEARN

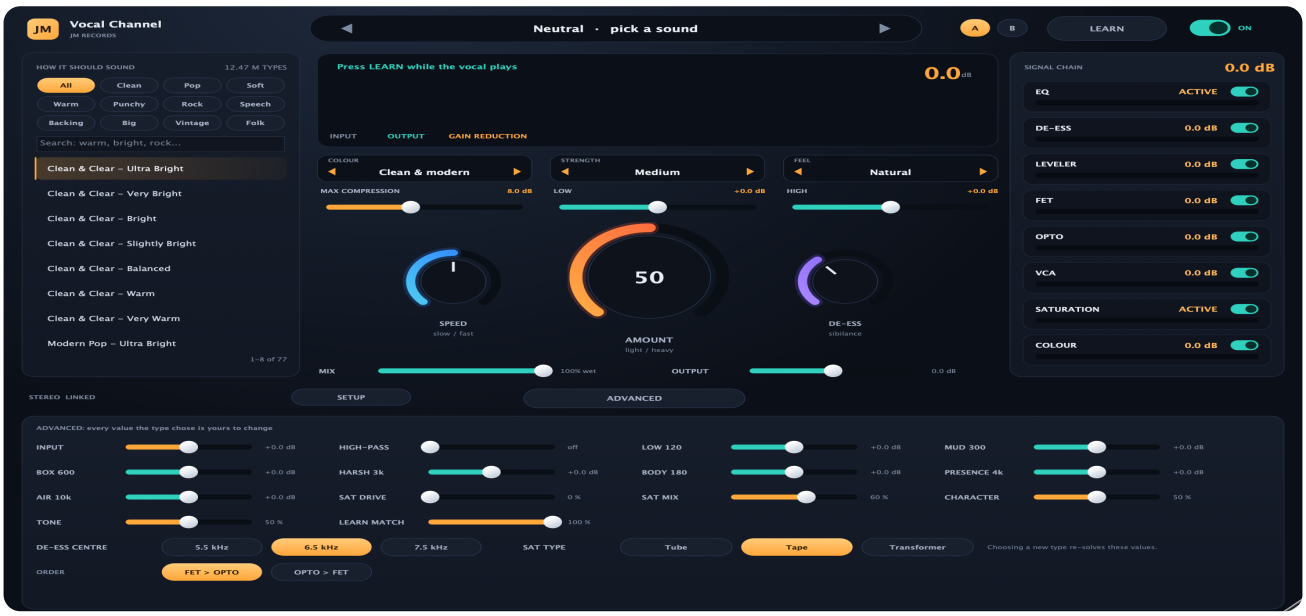
Part	What it does
VOICE	Soft / Breathy, Balanced, Powerful, Bright / Sibilant, Dark / Deep, Wide Dynamics.
RANGE	Very deep, Deep, Medium, High, Very high.
SOURCE	Studio mic, Stage / dynamic mic, Phone or laptop, Live recording, Echoey room, Old recording.
ROLE	Lead, Double, Backing, Ad-lib / Hook, Harmony / Choir.
Screen and LEARN	Level history and the coach message. Press LEARN, then play the vocal.
SKIP SETUP	Go to the main screen without LEARN.

Main screen



Main screen

Part	What it does
Top bar	Style name bar (◀ ▶), A/B compare, LEARN (learn again), ON / BYPASS.
VOCAL STYLE list	Category chips (ALL, Indian, Bengali, Pop, Rock, Urban, World, Speech, Utility), nine rows, mouse-wheel scrolling. Shows “Neutral · pick a style” until you pick one.
COLOUR, STRENGTH, FEEL	Three steppers that shape the chosen style (section 6).
MAX COMPRESSION	The most gain reduction the compressors may use, in dB (section 7).
LOW / HIGH	Small tone controls, –6 … +6 dB: LOW lifts or lowers body around 180 Hz, HIGH lifts or lowers the air above 10 kHz.
SPEED, AMOUNT, DE-ESS	The three knobs (section 7).
MIX / OUTPUT	Dry/wet blend and output level.
Signal chain cards (8)	EQ, DE-ESS, LEVELER, FET, OPTO, VCA, SATURATION, COLOUR with their gain reduction. Click a card to switch that stage off.
SETUP / ADVANCED	SETUP returns to the setup screen; ADVANCED opens the extra panel (section 9).



ADVANCED panel open

5. LEARN and the coach

The normal use: the vocal is already **recorded** on a mixer / DAW track and this plugin is on that track. Start playback of the recording (loop a part that has both soft and loud lines and some S sounds, for example verse into chorus), then press **LEARN**. The recorded signal passes through the plugin and it listens to it. (It works the same with a live singer.) The plugin listens to the raw input for 8 to 15 seconds of real voice. It stops by itself when the coach says **Enough**, or after 15 seconds of voice. At least about 2 seconds of voice are needed; with less, the result is ignored. Press LEARN again to cancel.

LEARN never changes the sound while it listens. When it is finished it stores a **voice profile** inside the session (saved with your project), sets the INPUT gain, the de-ess centre and MAX COMPRESSION, and switches on the corrective MATCH EQ. If a style is already chosen it is re-solved for the new profile. The audio excerpt LEARN keeps (up to 10 s of voiced audio) is not saved with the project: after reopening a project, press LEARN again if you want the compression fit and loudness matching to work for new style choices.

Coach message	Meaning and what to do
Keep going	Not enough information yet. Keep singing or speaking normally.
Add soft and loud	The loudness range heard so far is narrow. Sing a quiet phrase and a loud phrase so the plugin sees the real dynamics.
Add S sounds	Few sharp S / T / SH sounds were heard. Say a few such words so the de-esser centre can be chosen.
Enough	The plugin has what it needs; it will finish by itself.

MATCH EQ: why different recordings can sound alike. LEARN measures the tone of the recording in 24 narrow bands. The plugin compares it with a neutral reference vocal (the average of nine dry vocals) and builds up to **8 adaptive EQ bells** that correct the difference **before** the compressors: a room resonance gets

a narrow cut, a harsh or boxy area gets cut, a thin or dull recording gets lifted where possible. Then the chosen type (for example “Punchy”) is applied on top. So a recording made in a resonant room with a sharp tone and no low end and a clean studio recording both get pulled towards the same starting tone, and the same type then sounds much more alike. The LEARN MATCH slider sets how much of the correction is used.

What the profile contains:

Measured	Used for
Loud line	The INPUT trim is set so your loud passages reach the level the compressors were calibrated for.
Dynamic spread	A wide spread raises the amount of levelling; a narrow, already even voice gets less.
Noise floor / SNR	A noisy recording gets gentler downward shaping so noise is not lifted.
Rumble (below ~80 Hz)	Raises the HIGH-PASS frequency when low rumble is present.
Sibilance share	Chooses the DE-ESS CENTRE (5.5 / 6.5 / 7.5 kHz) and how strongly to de-ess.
24-band tone shape	Compared with the average of nine reference vocals. The MATCH EQ turns the difference into bells: resonances and harsh or boxy areas are cut, missing presence, air or body is lifted (MATCH never boosts below 300 Hz, only cuts there, so it cannot make a vocal boomy; boosts are limited to +3 dB at 300-500 Hz and about +9 dB above).

HONEST LIMITS

The “average” reference is the mean tone shape of only nine vocals. The coach thresholds and the band centres are assumptions. MATCH cannot restore what the recording does not contain (for example a low end that was filtered away completely); it can only lift what is there, within limits. In a test with the nine vocals deliberately damaged (resonant room, thin and sharp, boxy and dull), the damaged versions ended up about 43% closer in tone to the clean versions of the same vocal with MATCH than without. LEARN is a helpful starting measurement, not a perfect analysis.

6. Choosing the vocal type

The first four lists (VOICE, RANGE, SOURCE, ROLE) describe **your vocal** and are chosen on the setup screen. STYLE, COLOUR, STRENGTH and FEEL describe **the sound you want** and are chosen on the main screen. A **vocal type** is the combination of all eight choices. The style list has **60 styles**; the other seven lists have 5 to 6 entries each; 60 × 6 × 5 × 6 × 5 × 6 × 5 × 6 = **9,720,000 types**.

List	Choices and effect
STYLE (60)	The main character: the base amount, speed, tone, de-essing, order, EQ shape, saturation type and high-pass. See the full list below.
VOICE	Soft / Breathy, Balanced, Powerful, Bright / Sibilant, Dark / Deep, Wide Dynamics. Changes compression strength, de-essing and tone in the direction of the voice character.

RANGE	Very deep, Deep, Medium, High, Very high. Moves the high-pass and the low / mud / presence / air emphasis to fit the pitch range.
SOURCE	Studio mic, Stage / dynamic mic, Phone or laptop, Live recording, Echoey room, Old recording. Compensates for the recording chain (e.g. phone mic: more cleaning of box and harshness).
ROLE	Lead, Double, Backing, Ad-lib / Hook, Harmony / Choir. Backing and choir get more levelling and less presence, so they sit behind a lead.
COLOUR	Clean & modern, Warm & old-school, Bright & airy, Dark & moody, Gritty & driven, Lo-fi. Sets tone and saturation type and amount.
STRENGTH	Very light, Light, Medium, Strong, Very strong. Scales how far everything is pushed. Very light is the lightest, Very strong the heaviest.
FEEL	Natural, Soft, Punchy, Warm, Smooth, Dense. The mood on top of the style: Soft = gentler and less forward; Punchy = more compression bite, more presence and body; Warm = more body and tape-like colour; Smooth = calmer highs and more de-essing; Dense = heavily glued and thick.

The sound list has **77 sounds** = 11 families x 7 brightness steps. Every family has the same ladder from **Ultra Bright** (most air and presence, least mud) down through Very Bright, Bright, Slightly Bright and Balanced to Warm and Very Warm. Pick the family that matches the vocal, then the step that matches how bright you want it. Use the **family buttons** above the list to show one family, and the **search box** to find a sound by any word (for example *bright*, *warm*, *rock* or *soft very* - every word you type must match). Compression gets its own ladder: STRENGTH (Very light ... Very strong) and MAX COMPRESSION. Pick a sound, set STRENGTH, and most vocals need nothing from ADVANCED.

Category	Styles
Clean	Clean & Clear - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Pop	Modern Pop - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Soft	Soft & Intimate - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Warm	Warm & Smooth - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Punchy	Punchy & Loud - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Rock	Rock & Raw - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Speech	Speech - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Backing	Backing - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm

Big	Big & Emotional - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Vintage	Vintage - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm
Folk	Folk & Traditional - Ultra Bright, Very Bright, Bright, Slightly Bright, Balanced, Warm, Very Warm

GOOD TO KNOW

- Choosing a new type **re-solves the whole panel**. Manual ADVANCED tweaks are overwritten; your INPUT trim from LEARN is kept.
- The loudness of the result is kept close to the input: EQ and saturation include level compensation, so you can compare types fairly.

7. MAX COMPRESSION, knobs, LOW / HIGH, MIX, OUTPUT

MAX COMPRESSION is the safety limit against over-compression. LEARN proposes a value from your vocal (wide dynamics allow more, a noisy recording less; 4 to 12 dB, shown as for example 8.0 dB). Whenever you pick a style, the plugin runs the style on the vocal LEARN kept and, if the loud passages would be compressed harder than this limit (90th percentile of the total gain reduction), it lowers AMOUNT until they fit. Move the slider (2 … 16 dB) to allow more or less; AMOUNT is then re-fitted. It is a target measured on the learned vocal, not a brick wall: very rare peaks can briefly exceed it.

Control	What it does
AMOUNT	How hard the compressor chain works, within MAX COMPRESSION. The style and STRENGTH set it; you can fine-tune by hand.
SPEED	How fast the compressors react. Slow keeps punch and natural movement; fast catches peaks.
DE-ESS	Strength of the de-esser. Its centre (5.5 / 6.5 / 7.5 kHz) is chosen by LEARN, changeable in ADVANCED.
LOW	–6 … +6 dB at about 180 Hz, after the dynamics. A little more or less body. Works even before a style is picked.
HIGH	–6 … +6 dB high shelf above about 10 kHz. A little more or less air. Works even before a style is picked.
MIX	Blend between the unprocessed input (0%) and the processed sound (100%). The dry signal is time-aligned, so there is no comb filtering. Use it for parallel compression.
OUTPUT	Final level, –12 … +12 dB. When you pick a style the plugin sets it so the style has the same loudness as the corrected vocal; adjust it afterwards if you like.

INPUT (ADVANCED)	–15 … +15 dB trim in front of everything. LEARN sets it so the loud line meets the calibrated level.

Moving a knob or slider by hand does not change the chosen style; it adjusts the current result. If you move AMOUNT, SPEED or DE-ESS before you ever picked a style, the plugin leaves neutral mode and uses the panel values.

8. The signal path and the stage cards

Order: **INPUT → EQ (A) → DE-ESS → LEVELER → [FET → OPTO or OPTO → FET] → VCA catch → EQ (B) → SATURATION → COLOUR → MIX → OUTPUT.**

Stage	What it does
EQ (A + B)	A: high-pass and four bells that clean the source before dynamics (LOW 120 Hz, MUD 300 Hz, BOX 600 Hz, HARSH 3 kHz). B: sweetening after dynamics (BODY 180 Hz, PRESENCE 4 kHz, AIR shelf 10 kHz). Both are skipped completely when all gains are 0.
DE-ESS	Reduces S, T and SH sounds in one band, centred at 5.5, 6.5 or 7.5 kHz. Shows its reduction in dB.
LEVELER	Slow, gentle riding of the overall level, so phrases are about equally loud before the faster compressors.
FET	Fast, forward compressor. Catches peaks and adds energy.
OPTO	Smooth compressor with a slow, program-dependent release. Gives a natural, glued feel.
VCA	Fast safety compressor that catches leftover peaks.
SATURATION	Tube, Tape or Transformer style soft saturation (anti-aliased), DC-free, with unity small-signal gain and a dry/wet mix.
COLOUR	Tone shaping and harmonic colour, steered by CHARACTER and TONE.
ORDER	FET > OPTO: peaks first, then smooth. OPTO > FET: smooth first, then peak control. Try both; the difference is subtle but real.

Each card is also a switch: click it to turn that stage off (it is then bypassed with no click, with smoothing). A stage whose settings are neutral is skipped, so unused stages use no CPU.

9. ADVANCED mode

Click ADVANCED to open the panel under the main window. It shows what the selected style calculated, and lets you correct it.

Control	Range and use

INPUT	–15 … +15 dB. Level into the chain (set by LEARN).
HIGH-PASS	0 … 300 Hz (0 = off). Removes rumble and handling noise.
LOW 120	–9 … +9 dB bell at 120 Hz. Cut for boom, lift for weight.
MUD 300	–9 … +9 dB at 300 Hz. Usually a cut to clean a muddy voice.
BOX 600	–9 … +9 dB at 600 Hz. A cut removes the “boxy” sound of small rooms.
HARSH 3k	–9 … +9 dB at 3 kHz. A small cut tames harsh edges.
BODY 180	–9 … +9 dB at 180 Hz, after dynamics: warmth and size. The main screen LOW slider adds to this.
PRESENCE 4k	–9 … +9 dB at 4 kHz, after dynamics: clarity and forwardness.
AIR 10k	–9 … +9 dB high shelf at 10 kHz. The main screen HIGH slider adds to this.
SAT DRIVE / SAT MIX	0 … 100%. How hard the saturator is driven (styles never go above 60%) and the saturated / clean blend.
CHARACTER	0 … 100%. How much of the colourful behaviour of the FET and opto stages is used.
TONE	0 … 100%. Tilt of the colour stage: darker to the left, brighter to the right.
LEARN MATCH	0 … 100%. How much of LEARN's corrective EQ (resonance cuts, tone matching) is used. 0% switches it off.
DE-ESS CENTRE	5.5 / 6.5 / 7.5 kHz. LEARN picks the one where the S sounds live.
SAT TYPE	Tube (warm), Tape (soft, rounded top), Transformer (low-end weight).
ORDER	FET > OPTO: peaks first, then smooth. OPTO > FET: smooth first, then peak control.

GUARD RAILS

Whatever you choose, a style never uses more than about 12 dB of EQ in total (no single band beyond ±6 dB), saturation drive stays at 60% or less, AMOUNT stays at 85% or less, and the compression stays within MAX COMPRESSION. Manual sliders in ADVANCED can go further if you want.

WORKFLOW

Change one slider at a time, use A/B to compare. If you pick another style afterwards, the sliders are re-solved, so choose the style first, then tweak.

10. A/B, bypass, presets, sessions

Item	Behaviour

A/B	Two memory slots for the whole panel: set up A, copy to B, change B, then switch to compare.
BYPASS / ON	Smoothly crossfades between processed and unprocessed (no click). The bypassed sound is the raw input at the same position in time.
Programs	In the DAW's preset menu: “Init” plus the 77 sounds. Choosing one is the same as choosing the sound in the list.
Automation	All 42 parameters (including the eight type lists) are host parameters. Note: automating a type list does not re-solve the other settings; use the plugin window to pick a type.
Session save	The panel and the LEARN profile are saved with the DAW project, so it recalls exactly.
Stereo	The two channels are always linked, so the image does not move. Mono works too.

11. Trial key (beta)

This version needs a **licence key**. A **trial key** (sent by e-mail after you register) works for the number of days written in the e-mail, counted from the day it was issued (usually 30). A **permanent key** never expires. Both are entered the same way and are tied to your e-mail address.

Step / item	What to do or know
Enter the key	Open the plug-in. At the bottom right the button reads ENTER YOUR LICENCE KEY. Click it, paste the key (it starts with JMVC1.) and press Activate. If the key is on the clipboard it is already filled in.
While the trial runs	The button shows TRIAL: N DAYS LEFT. Everything works normally. With a permanent key it shows LICENSED.
Buying the plug-in	When you buy, you receive a permanent key by e-mail. Enter it the same way: it replaces the trial key, and your projects and settings stay exactly as they were.
Without a valid key	The plug-in does not change the sound at all: your audio passes through unprocessed. The button says what is wrong: ENTER YOUR LICENCE KEY, TRIAL ENDED - ENTER LICENCE KEY, KEY NOT VALID - TRY AGAIN or CHECK THE COMPUTER DATE.
One key for every project	The key is stored once per computer user (macOS: ~/Library/Application Support/JM Records/Vocal Channel/licence.txt; Windows: C:\Users\<name>\AppData\Roaming\JM Records\Vocal Channel\licence.txt). All projects and all DAWs use it; you enter it only once.
Computer date	The plug-in checks the date. Turning the computer clock back to extend the trial is detected and switches the plug-in to pass-through.
Keep your projects	When the trial ends your projects, presets and settings are kept. After a new key is entered everything sounds exactly as before.

Feedback	Please report what you liked, what was too bright, dull, harsh or muddy, which sound and which vocal, and your DAW. That is the purpose of the trial.

12. Troubleshooting

Problem	Try this
Sound does not change at all	Check the button at the bottom right: it needs a valid trial key (section 11). Without one the plug-in passes the audio through unprocessed.
Plugin not in the DAW list	Check the folders in section 2 (on Windows the whole .vst3 folder must be inside C:\Program Files\Common Files\VST3). macOS: run killall -9 AudioComponentRegistrar. Then rescan in the DAW. Some DAWs need a restart.
LEARN gives a strange setting	The voice was too short, too quiet or had a different mic. Press LEARN again and sing a full phrase: soft, loud, and a few S sounds.
Too much compression	Lower MAX COMPRESSION (main screen), lower STRENGTH, or turn AMOUNT down, or switch OPTO / FET card off to hear the difference.
S sounds are sharp	Raise DE-ESS, and try the other DE-ESS CENTRE in ADVANCED.
Sound is dull	Increase PRESENCE or AIR in ADVANCED or choose a brighter step in the sound list (Bright, Very Bright, Ultra Bright).
Sound is boomy or thin	Lower the main-screen LOW slider, or cut LOW 120 / BODY 180 and MUD 300 in ADVANCED; raise them if the sound is thin. MATCH never boosts below 300 Hz.
My tweaks vanished	Picking a new type or LEARN re-solves the panel. Choose the type first, then tweak.
Noise gets louder	Lower AMOUNT/STRENGTH; compressors lift quiet noise. Clean the source or use a gate before the plugin.

JM Records · JM Vocal Channel (working name; the final product name is still open). Not affiliated with, and no claim of matching, any hardware maker or product.